

Assemble a Picture Frame SOP

One episode builds one picture frame. The frame starts face down on the work board with its backing board held in by two tabs. The episode opens the tabs, puts the glass and the photo in, closes the tabs, attaches the stand, stands the frame up, and wipes the glass.

The order never changes: open both tabs, take the backing board out, put the glass in, lay the photo on the glass, put the backing board back, close both tabs, attach the stand, set the frame upright, then wipe the glass. Nothing goes into the frame well while a tab is still over it.

The right gripper does the main work: the stand, the lift that stands the frame up, and the wipe. The left gripper supports: it works the left tab, moves the backing board, and holds the frame steady. The glass and the photo go in with whichever gripper is on the parts shelf's side. Neither gripper reaches across the table into the other one's zone.

The frame is never lifted off the work board. Every flat part is carried level and lowered into the frame well until it lies flat, never dropped in from above and never dragged. No gripper ever closes on the face of the glass.

The table is set up in one of three ways. Only the parts shelf, with the glass and the photo on it, moves; the work board, the backing rest, the stand spot, and the cloth spot are in the same place in all three.

- **Config L:** the parts shelf is at the back-left.
- **Config M:** the parts shelf is at the front-center, in front of the work board.
- **Config R:** the parts shelf is at the right, between the stand spot and the cloth spot.

Where a step depends on the setup it says so on an **IF** line — look at the table and follow the line that matches.

What stays constant across all sessions:

- **Start position:** the parts shelf starts at the back-left (**Config L**), the front-center (**Config M**) or the right (**Config R**). One config per episode, chosen before recording and never changed mid-episode.
- **Same-side rule:** the gripper on the parts shelf's side puts the glass and the photo in — the left gripper in Config L and M, the right gripper in Config R. No arm reaches across the table.
- **Fixed roles:** the left gripper works the left tab, carries the backing board, and holds the frame; the right gripper works the right tab, attaches the stand, stands the frame up, and wipes. These do not change with the config.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole table: the work board at the center with the frame face down on it, the backing rest at the left, the parts shelf in the start zone for this episode's config, the stand spot at the back right, and the cloth spot at the front right.
3. The back of the frame is visible from above, so both tabs, the backing board, and the frame well can be seen.
4. The work board is visible from the front, so the frame can be seen standing upright and the whole glass face can be seen during the wipe.
5. Both arms are at home with grippers open.
6. The table is clean, dry, and bare apart from the work board, the backing rest, the parts shelf, the stand spot, and the cloth spot.
7. The left arm reaches the backing rest, the left tab, the left side rail of the frame, and (Config L and M) the parts shelf. The right arm reaches the stand spot, the cloth spot, the right tab, the whole frame, and (Config R) the parts shelf.

Materials checklist

1. One **picture frame** lies face down on the **work board** at the center, back facing up, with the **backing board** in place and both **tabs** closed over it.
2. The frame sits square on the work board with its top rail toward the back of the table. The top rail hangs past the back edge of the board, leaving a **lift gap** under it.
3. The frame is light enough that one gripper can tip it up on its bottom rail, and heavy enough to stay put while a part is lowered into it.
4. The frame has two **tabs**, one at the left edge of the back and one at the right edge. Each tab swings flat and is wide enough for a gripper to pinch.
5. The backing board has a **pull tab** standing up at its center and a **stand slot** cut near the edge that faces the back of the table.
6. One **parts shelf**, a low flat block, stands in the start zone for this episode's config. The **glass** lies flat on its back half. The **photo** lies flat on its front half, picture side down.
 - **Config L:** back-left.
 - **Config M:** front-center, in front of the work board.

- **Config R:** right, between the stand spot and the cloth spot.
7. The glass and the photo both hang past one side of the shelf, so a gripper can take an **edge grip** on each one: the left edge past the left side (Config M and R), or the right edge past the right side (Config L).
 8. The glass is one clear pane, clean, dry, and smooth at the edges. It drops into the frame well with an easy fit and does not have to be pressed in.
 9. The photo is one print, cut a little smaller than the frame well, flat and uncreased.
 10. One **stand** sits on the **stand spot** at the back right: a leg with a flat **tongue** at one end. The body of the leg is thick enough for a gripper to hold.
 11. One dry soft **cloth** sits folded into a small square on the **cloth spot** at the front right, small enough for one gripper to hold.
 12. The **backing rest** at the left is empty. It is a flat block a little larger than the backing board.

Workspace layout

- **Work board:** center, holding the frame face down with the top rail over the lift gap.
- **Backing rest:** left, empty, where the backing board waits.
- **Parts shelf (start zone):** back-left (Config L), front-center in front of the work board (Config M), or right between the stand spot and the cloth spot (Config R), holding the glass on its back half and the photo on its front half.
- **Stand spot:** back right, holding the stand.
- **Cloth spot:** front right, holding the folded cloth.

Everything the right gripper carries stays on the right and the center. The left gripper works the left of the table, the front-center, and the center. Neither one crosses to the other side.

Vocabulary

- **Arm assignment:** the right gripper works the right tab, attaches the stand, stands the frame up, and wipes the glass, and puts the glass and the photo in when the parts shelf is at the right (Config R). The left gripper works the left tab, carries the backing board, and holds the frame steady, and puts the glass and the photo in when the parts shelf is at the back-left or the front-center (Config L and M). Neither gripper takes the other one's work.
- **Start zone:** where the parts shelf stands at the start of the episode — back-left (**Config L**), front-center (**Config M**), or right (**Config R**). One per episode, chosen before recording and never changed mid-episode.
- **Frame well:** the shallow space inside the back of the frame where the glass, the photo, and the backing board sit. Its edge all the way round is the **frame lip**.
- **Tab:** one of the two flat swivel tabs on the back of the frame. **Open** means the tab points outward and nothing of it lies over the frame well. **Closed** means the tab lies flat across the

backing board.

- **Backing board:** the stiff panel that closes the back of the frame. It carries the pull tab and the stand slot.
- **Pull tab:** the stiff card tab standing up at the center of the backing board. It is the only place a gripper takes hold of that board.
- **Stand slot:** the cut in the backing board that the stand tongue slides into.
- **Edge grip:** the gripper closes on the edge of a flat part that hangs past the shelf, one finger above it and one below, so the part stays level. Used for the glass and the photo.
- **Lying flat:** the part is down in the frame well, resting on what is under it, with no corner riding up on the frame lip. The gripper opens only once this is true.
- **Frame hold:** the left gripper closed on a rail of the frame and pressing it against the work board, so the frame cannot slide, turn, or tip while the right gripper works. The hold stays on for the whole step.
- **Lift gap:** the space under the top rail where it hangs past the back edge of the work board. It is where the right gripper gets under the frame to stand it up.
- **Upright:** the frame standing on its bottom rail with the front facing the front edge and the stand leg foot resting on the work board behind it.
- **Down stroke:** the right gripper presses the cloth flat on the glass at the top and draws it straight down to the bottom of the glass in one go, then lifts the cloth clear.
- **Release point:** over the frame well, over the backing rest, or over the cloth spot. A gripper opens nowhere else.

Steps

Steps 1 to 3 fill the frame and close it. Step 4 attaches the stand. Step 5 stands the frame up. Step 6 wipes the glass. Step 7 ends the episode. Only Step 2 depends on the config: in Config L and M the **left gripper** takes the glass and the photo from the parts shelf, in Config R the **right gripper** does. Every other line is the same in all three.

Step 1: Open the tabs and take the backing board out

Goal: both tabs open, the frame well empty, and the backing board lying on the backing rest.

- The **left gripper** pinches the **left tab** and swings it outward until it is flat and clear of the frame well, then releases.
- The **right gripper** does the same to the **right tab**.
- The **left gripper** closes on the **pull tab**, lifts the backing board straight up out of the frame well, carries it level to the **backing rest**, lays it flat with the pull tab up, and releases.

Open both tabs before the board moves. Never pry the board out with a tab still over it, and never take hold of the board anywhere but the pull tab.

Check: both tabs point outward and clear the frame well, the frame well is empty, the frame has not moved on the work board, and the backing board lies flat on the backing rest with the pull tab up.

Step 2: Put the glass and the photo in

Goal: the glass lying flat in the frame well and the photo lying flat on the glass, blank side up.

2.1 Put the glass in

- **IF the parts shelf is at the back-left (Config L):** the **left gripper** takes an **edge grip** on the **glass** at the parts shelf and lifts it straight up.
- **IF the parts shelf is at the front-center (Config M):** the **left gripper** takes an **edge grip** on the **glass** at the parts shelf and lifts it straight up.
- **IF the parts shelf is at the right (Config R):** the **right gripper** takes an **edge grip** on the **glass** at the parts shelf and lifts it straight up.

Then, in all three:

- Carry it level to the frame, lower it into the **frame well**, and release once it is **lying flat**.

2.2 Put the photo in

- **IF Config L or M:** the **left gripper** takes an **edge grip** on the **photo** and lifts it straight up.
IF Config R: the **right gripper** does the same.
- Carry it level to the frame, lower it onto the glass, and release once it is **lying flat**.

Glass first, then the photo. Never close on the face of the glass. Never let go of a part above the frame, and never slide a part across the work board into place.

Check: the glass lies flat in the frame well, the photo lies on the glass with the blank side up, no corner of either part rides up on the frame lip, the parts shelf is empty, and the frame has not moved.

Step 3: Put the backing board back and close the tabs

Goal: the backing board flat in the frame well and both tabs closed over it.

- The **left gripper** closes on the **pull tab**, lifts the backing board straight up off the backing rest, carries it level to the frame, lowers it onto the photo, and releases once it is **lying flat**. The **stand slot** ends up near the top rail.
- The **left gripper** pinches the **left tab** and swings it in until it lies flat across the backing board, then releases.
- The **right gripper** does the same to the **right tab**.

Put the board in the right way round the first time. Never turn it over on the frame, and never close a tab over the stand slot.

Check: the backing board lies flat in the frame well with no corner on the frame lip, both tabs lie flat across the board, the stand slot is near the top rail and is clear of both tabs, and the backing rest is empty.

Step 4: Attach the stand

Goal: the stand tongue all the way into the stand slot and the leg lying flat on the backing board.

- The **left gripper** takes a **frame hold** on the **left side rail**. The hold stays on until the end of the step.
- The **right gripper** closes on the body of the **stand** at the stand spot, lifts it straight up, carries it level to the frame, and lines the **tongue** up with the **stand slot**.
- The **right gripper** pushes the tongue straight into the slot until the leg lies flat on the backing board pointing toward the bottom rail, then releases.
- The **left gripper** releases the side rail.

Push straight in. Never lever the tongue in sideways, and never rest the leg on the board without putting the tongue in the slot.

Check: the tongue is all the way into the stand slot, the leg lies flat pointing toward the bottom rail, the stand stays put once the right gripper opens, the frame has not moved on the work board, and the stand spot is empty.

Retry: if the tongue will not go in, the **right gripper** pulls the stand straight back out, lines it up again, and pushes again.

Step 5: Set the frame upright

Goal: the frame standing **upright** on the work board, front facing the front edge, resting on its stand leg.

- The **left gripper** takes a **frame hold** on the **bottom rail**. This is the pivot, and it stays on until the frame is standing.
- The **right gripper** closes its fingers, slides them into the **lift gap** under the top rail, and lifts the top rail up and toward the front edge, so the frame turns up on its bottom rail.
- Keep lifting until the frame stands and the stand leg swings down and its foot lands on the work board behind it.
- The **right gripper** comes out from under the frame and moves clear. The **left gripper** then releases the bottom rail.

The bottom rail stays where it is. Never lift the frame off the work board, never pull it toward the front edge, and never leave it leaning on a gripper.

Check: the frame stands upright on the work board, the front faces the front edge, the stand leg foot rests on the board behind the frame, the frame does not rock or slide once both grippers are clear, and nothing has come out of the frame.

Retry: if the frame falls back down face down on the work board, the **right gripper** lifts the top rail again from the lift gap.

Step 6: Wipe the glass

Goal: the whole glass face wiped in three down strokes with the frame still standing.

- The **left gripper** closes on the **left side rail** of the standing frame and holds it steady. The hold stays on until the end of the step.
- The **right gripper** closes on the **cloth** at the cloth spot and lifts it straight up.
- The **right gripper** makes one **down stroke** at the left of the glass, then one down the middle, then one at the right, so the three strokes cover the whole glass face.
- The **right gripper** carries the cloth back to the **cloth spot**, lays it down, and releases. The **left gripper** then releases the side rail.

Wipe downward only. Never wipe up or across, never press hard enough to push the frame off its leg, and never wipe the frame rails.

Check: three down strokes were made, left, then middle, then right, covering the whole glass face, the frame still stands in its place with the leg foot on the board, and the cloth lies back on the cloth spot.

Step 7: Confirm the result and end the episode

- Confirm the frame stands upright with the front facing the front edge, the photo shows through the glass, both tabs are closed, the stand leg foot is on the work board, and the backing rest, the parts shelf, and the stand spot are all empty with the cloth on the cloth spot.
- Return both arms home, clear of the frame and the work board, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Lay the frame back face down on the work board, square, with the top rail hanging over the back edge so the lift gap is there again.
2. Pull the stand out of the stand slot and put it back on the stand spot at the back right.
3. Open both tabs, lift the backing board out, and take the photo and the glass out of the frame well.
4. Put the parts shelf in the start zone for the next episode's config — back-left (Config L), front-center in front of the work board (Config M), or right between the stand spot and the

cloth spot (Config R).

5. Wipe both faces of the glass, dry it, and lay it flat on the back half of the parts shelf with its left edge (Config M and R) or right edge (Config L) hanging past the side of the shelf.
6. Lay the photo picture side down on the front half of the parts shelf, the same edge hanging past the side of the shelf.
7. Put the backing board back into the empty frame well with the pull tab up and the stand slot near the top rail, then close both tabs over it. The backing rest stays empty.
8. Shake the cloth out, fold it back into a small square, and put it on the cloth spot. Swap in a dry cloth if it is damp or dirty.
9. Wipe the table and the work board and confirm both are clean and dry. A wet board lets the frame slide.
10. Replace a cracked or chipped glass, a creased or torn photo, a tab that will not stay closed, a split backing board, a pull tab that has come loose, and a stand whose tongue no longer grips the slot.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Note on the start position: the violations below were written for Config R (parts shelf at the right, the right gripper putting the glass and the photo in). The pickup and arm-role cues will be rewritten later to cover all three start positions; they are left as they are for now. Until then, anything that does not match the episode's config goes under **Config misaligned**.

Violation: Config misaligned

- **Visible cue:** what the operator does does not match the config on the table — the parts shelf is not in the start zone for the config; a gripper reaches across the table for the glass or the photo; or the wrong IF line is followed.
- **SOP rule broken:** the start position and the same-side rule (the gripper on the parts shelf's side takes the glass and the photo, the left gripper in Config L and M and the right gripper in

Config R; no arm reaches across the table; the IF line followed is the one for the config on the table).

- **Coaching note:** look where the parts shelf is before the first reach, then follow that config's IF lines through Step 2.

Violation: Wrong order

- **Visible cue:** the photo goes into the frame well before the glass, the backing board goes back before the photo is in, a tab is closed before the backing board is in, the stand goes on before both tabs are closed, or the glass is wiped before the frame is standing.
- **SOP rule broken:** Steps 1 to 6, glass, then photo, then backing board, then both tabs closed, then the stand, then upright, then the wipe.
- **Coaching note:** run the frame in one direction. Nothing goes in until what belongs under it is already flat.

Violation: Arms swapped roles

- **Visible cue:** the left gripper picks the glass, the photo, the stand, or the cloth, works the right tab, or lifts the top rail, or the right gripper carries the backing board or works the left tab.
- **SOP rule broken:** Steps 1 to 6, the right gripper places the glass, the photo, and the stand, stands the frame up, and wipes, and the left gripper works the left tab, carries the backing board, and holds the frame.
- **Coaching note:** the left gripper holds and passes the board. The fine work is the right gripper's.

Violation: Tab forced or not opened

- **Visible cue:** the backing board is lifted or pried while a tab still lies over the frame well, a tab is bent or dragged instead of swung flat, or a tab is left half turned so part of it still covers the well.
- **SOP rule broken:** Step 1, both tabs swing outward flat and clear of the frame well before the backing board moves.
- **Coaching note:** swing both tabs right out first. A board that needs force means a tab is still over it.

Violation: Backing board mishandled

- **Visible cue:** a gripper takes the backing board by an edge or a corner instead of the pull tab, the board is dragged across the frame or the table, it is dropped onto the rest or into the frame, it lands pull tab down, or it is left hanging off the backing rest.
- **SOP rule broken:** Steps 1 and 3, the left gripper holds the pull tab, lifts the board straight up, carries it level, and lays it flat.
- **Coaching note:** always by the pull tab, always straight up and straight down.

Violation: Glass or photo handled wrong

- **Visible cue:** the right gripper closes on the face of the glass instead of taking an edge grip, a part is let go above the frame and falls in, a part is tilted or dragged into the well, or the photo goes in picture side up.
- **SOP rule broken:** Step 2, edge grip on the glass and the photo, carried level, lowered into the frame well, photo blank side up.
- **Coaching note:** grip the edge that hangs off the shelf, keep the part level, and take it all the way down before you open.

Violation: Part not lying flat

- **Visible cue:** at the end of Step 2 or Step 3 the glass, the photo, or the backing board has a corner riding up on the frame lip, sits on top of the lip instead of down in the well, or the photo is bent under the backing board.
- **SOP rule broken:** Steps 2 and 3, every part is lowered until it is lying flat in the frame well before the gripper opens.
- **Coaching note:** watch the corners on the way down. If one catches the lip, lift and lower again.

Violation: Tabs left open or over the slot

- **Visible cue:** at the end of Step 3 a tab still points outward, a tab lies only part way across the backing board, or a tab lies over the stand slot.
- **SOP rule broken:** Step 3, both tabs swing in and lie flat across the backing board with the stand slot left clear.
- **Coaching note:** both tabs flat, and keep the slot clear for the stand.

Violation: Stand not seated

- **Visible cue:** the tongue goes only part way into the stand slot, the stand comes free when the right gripper opens, the stand is levered in sideways, the leg is laid on the backing board with the tongue never in the slot, or the right gripper pushes with no frame hold on.
- **SOP rule broken:** Step 4, the left gripper holds the frame while the right gripper pushes the tongue straight into the slot until the leg lies flat.
- **Coaching note:** hold the frame first, line the tongue up, then push straight in.

Violation: Frame lifted or tipped wrong

- **Visible cue:** the frame is lifted off the work board to stand it up, the right gripper grips the top rail instead of pushing up from the lift gap, the left gripper comes off the bottom rail while the frame is going up, the frame slides toward the front edge, or the frame is left leaning on a gripper.

- **SOP rule broken:** Step 5, the left gripper pins the bottom rail as the pivot and the right gripper lifts the top rail from the lift gap until the stand leg lands on the board.
- **Coaching note:** the bottom rail never moves. Turn the frame up on it.

Violation: Wipe not done as prescribed

- **Visible cue:** fewer or more than three down strokes, a stroke that goes up or across, the wipe runs with the left gripper off the side rail, the wipe pushes the frame off its leg or knocks it over, the cloth wipes the frame rails, or the cloth is left on the frame or the work board.
- **SOP rule broken:** Step 6, the left gripper holds the side rail while the right gripper makes three down strokes, left, then middle, then right, and returns the cloth to the cloth spot.
- **Coaching note:** hold the frame, wipe downward only, three strokes, then park the cloth.

Violation: Dropped, collided, or knocked something over

- **Visible cue:** the glass, the photo, the backing board, the stand, or the cloth falls to the table or the floor, the standing frame is knocked over, the work board, the parts shelf, or the backing rest is struck and shifts, or the two arms strike each other.
- **SOP rule broken:** Steps 1 to 6, carry each part level on a clear path and open only over a release point.
- **Coaching note:** check the path and the landing spot before you move.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with the frame still face down, a part still on the parts shelf, the backing board still on the backing rest, the stand still on the stand spot, the cloth still in a gripper, an arm away from home, or an arm still over the frame.
- **SOP rule broken:** Step 7, confirm the standing frame, the closed tabs, the stand leg, and the cleared zones, return both arms home clear of the frame, then stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Defective glass:** a pane that is cracked, chipped, or too tight for the frame well under a correct lowering. Replace it before the next episode.
- **Defective photo:** a print that is curled, creased, or torn so it will not lie flat under a correct placement. Replace it before the next episode.

- **Defective frame:** a tab that will not stay closed, a frame well that is out of shape, or a frame so light it lifts with a correct tab pinch. Replace it before the next episode.
- **Defective backing board:** a split board, or a pull tab that pulls loose under a correct grip. Replace it before the next episode.
- **Stand fault:** a tongue that no longer grips the slot, or a leg that folds under the standing frame. Replace it before the next episode.
- **The frame, the parts shelf, or the backing rest slides** under a correct hold because the table or the work board is wet. Wipe and dry both before the next episode.

Annotation subtasks (from SOP)

1. Open the left tab with the left gripper and the right tab with the right gripper
2. Lift the backing board out and lay it on the backing rest
3. Put the glass into the frame well
4. Lay the photo on the glass
5. Put the backing board back into the frame well
6. Close both tabs over the backing board
7. Hold the frame with the left gripper and push the stand tongue into the slot
8. Turn the frame up on its bottom rail onto its stand
9. Wipe the glass in three down strokes
10. Return the cloth to the cloth spot
11. Confirm the result and end the episode